

PROGRAM CHARTER

Municipal Water Infrastructure Capital Improvement Program

Project Controls & Risk Management System — Portfolio Case Study

Program Name	HWSD Municipal Water Capital Improvement Program
Sponsoring Utility	Harborview Water & Sewer Department (HWSD) — fictional, simulated
Business Need	Aging drinking-water infrastructure, rising failure risk on critical assets, regulatory compliance obligations, and planned service-area growth require a coordinated, multi-year capital delivery program rather than isolated, uncoordinated projects.
Program Objective	Deliver 12 coordinated water-infrastructure projects over 36 months within the approved \$96,000,000 total program authorization (\$88,500,000 in project budgets plus a \$7,500,000 contingency reserve), while meeting scope, schedule, quality, safety, and service-continuity objectives.
Success Measures	Cost performance (CPI), schedule performance (SPI), milestone adherence, risk exposure trend, change-order rate, contingency consumption, and reporting timeliness.
In Scope	Program controls, planning, construction/implementation tracking, risk and change management, vendor/contractor coordination, and monthly executive reporting.
Out of Scope	Detailed hydraulic design, civil/structural design calculations, stamped engineering drawings, and real municipal procurement actions.
Governance	Executive Sponsor → Program Manager → Project Managers → Discipline Leads / Contractors → Controls Analyst.
Reporting Cadence	Monthly program review; weekly project-level status updates.
Project BAC Total (P01–P12)	\$88,500,000 — the sum of the 12 projects' scoped, baselined work. This is the figure used for all Earned Value Management (EVM) rollups.
Program Contingency Reserve	\$7,500,000 — unallocated funds held at the program level, controlled by the Program Manager, for known-unknown risks within the program's risk profile. Not assigned to any project; released only through an approved change request.
Total Program Authorization	\$96,000,000 (Project BAC Total + Program Contingency Reserve) — the total amount authorized/appropriated for the program.
Program Duration	36 months (January 2025 – December 2027).
Key Assumption	All financial, schedule, risk, and performance data in this program is synthetic and intended solely for portfolio demonstration. No real utility, contractor, or individual is represented.

Disclaimer

This program charter, and all supporting scope, schedule, budget, risk, and performance data in this repository, are simulated for the purpose of demonstrating industrial engineering and project-controls methods. Harborview Water & Sewer Department (HWSD) is a fictional utility. This document does not represent an actual municipal capital program and does not constitute licensed civil engineering work.

Prepared as part of a project-controls portfolio project | Status date: March 31, 2026 (Month 15 of 36)